4.2.1. Month with the Highest Total Sales 02:38

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: `Date`, `Product`, `Quantity`, `Price`, and `City`.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases +

Explorer | monthForS... | Submit | Debugger

```python
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)
# Calculate the total sales for each transaction (Quantity * Price)
df['Total Sales'] = df['Quantity'] * df['Price']
# Convert the 'Date' column to datetime objects to extract month information
df['Date'] = pd.to_datetime(df['Date'])
df['Month'] = df['Date'].dt.to_period('M')
# Group by Month and sum the Total Sales
monthly_sales = df.groupby('Month')['Total Sales'].sum()
# Find the month with the highest total sales
best_month = monthly_sales.idxmax()
highest_sales = monthly_sales.max()

print(f"Best month: {best_month}")
print(f"Total sales: ${highest_sales:.2f}")
```

CODETANTRA

4.2.2. Best Selling Product

01:21

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: `Date`, `Product`, `Quantity`, `Price`, and `City`.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases +

Explorer

monthForS...

Submit

Debugger

```python
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)
product_quantity = df.groupby('Product')['Quantity'].sum()

# Find the product with the highest total quantity sold
best_product = product_quantity.idxmax()
highest_quantity = product_quantity.max()

# Display the result
print(f"Best selling product: {best_product}")
print(f"Total quantity sold: {highest_quantity}")

```

CODETANTRA

4.2.3. City that Sold the Most Products

00:58

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: `Date`, `Product`, `Quantity`, `Price`, and `City`.
- Group the data by `City` and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases +

Explorer | monthForS... | Submit | Debugger

```python
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)

# write the code..
city_sales = df.groupby('City')['Quantity'].sum()
best_city = city_sales.idxmax()
# Display the result
print(f"City sold the most products: {best_city}")

```

C☀DETANTRA

4.2.4. Most Frequently Sold Product Pairs 04:06

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Explanation:

Transactions:

- **2025-01-01:** Product A, Product B
- **2025-01-02:** Product A, Product C
- **2025-01-03:** Product B, Product A
- **2025-01-04:** Product C, Product B
- **2025-01-05:** Product A, Product C

Now, let's count how often the pairs of products appear together:

- **Product A and Product B**: Appear in transactions on 2025-01-01 and 2025-01-03.
- **Product A and Product C**: Appear in transactions on 2025-01-02 and 2025-01-05.
- **Product B and Product C**: Appears in transactions on 2025-01-04.

Most Frequent Product Combinations:

- **Product A and Product B** (2 times)
- **Product A and Product C** (2 times)

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases +

Explorer | frequently... | Submit | Debugger

```python
import pandas as pd
from itertools import combinations
from collections import Counter

# Prompt user to input the file name
file_name = input()

# Read data from the specified CSV file
df = pd.read_csv(file_name)
daily_transactions = df.groupby('Date')['Product'].apply(list)
pair_counts = Counter()
for products in daily_transactions:
    products = sorted(products)
    pairs = list(combinations(products, 2))
    pair_counts.update(pairs)
if pair_counts:
    max_count = max(pair_counts.values())
    for pair, count in pair_counts.items():
        if count == max_count:
            print(f"{pair[0]} and {pair[1]}: {count} times")
else:
    print("No product pairs found.")
```

CODETANTRA

4.2.5. Titanic Dataset Analysis and Data Clea... 05:59

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using `.info()`).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.
8. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
9. Drop the 'Cabin' column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ti
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thay
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",fe
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.0
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.86
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,34990
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,
```

Note: Refer to the visible test case for better reference.

Sample Test Cases +

Explorer | titanicData... | Submit | Debugger

```python
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# 1. Display the first 5 rows of the dataset
print(data.head())
# 2. Display the last 5 rows of the dataset
print(data.tail())
# 3. Get the shape of the dataset
print(data.shape)
# 4. Get a summary of the dataset (info)
print(data.info())
# 5. Get basic statistics of the dataset
print(data.describe())
# 6. Check for missing values
print(data.isnull().sum())
# 7. Fill missing values in the 'Age' column with the median age
data['Age'].fillna(data['Age'].median(), inplace = True)
# 8. Fill missing values in the 'Embarked' column with the mode
data['Embarked'].fillna(data['Embarked'].mode()[0],inplace=True)
# 9. Drop the 'Cabin' column due to many missing values
data.drop(columns='Cabin',inplace = True)
# 10. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch'
data["FamilySize"] = data['SibSp']+data['Parch']
```

CODETANTRA

4.2.6. Titanic Dataset Analysis and Data Clea... 04:29

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
2. Convert the 'Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the 'Embarked' column, dropping the first category.
4. Get the mean age of passengers.
5. Get the median fare of passengers.
6. Get the number of passengers by class.
7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ti
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thay
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",fe
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.0
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.86
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,34990
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,
```

Note: Refer to the visible test case for better reference.

Sample Test Cases +

Explorer

titanicData...

Submit

Debugger

```python
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']

# 1. Create a new column 'IsAlone' (1 if alone, 0 otherwise)


# 2. Convert 'Sex' to numeric (male: 0, female: 1)
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
# 3. One-hot encode the 'Embarked' column
data = pd.get_dummies(data, columns=['Embarked'])
# 4. Get the mean age of passengers
mean_age = data['Age'].mean()
print( mean_age)
# 5. Get the median fare of passengers
median_fare = data['Fare'].median()
print( median_fare)
# 6. Get the number of passengers by class
print( data['Pclass'].value_counts())
# 7. Get the number of passengers by gender
print( data['Sex'].value_counts())
# 8. Get the number of passengers by survival status
print( data['Survived'].value_counts())
# 9. Calculate the survival rate
survival_rate = data['Survived'].mean()
print(format(survival_rate))
# 10. Calculate the survival rate by gender
survival_by_gender = data.groupby('Sex')['Survived'].mean()
print( survival_by_gender)
```

CODETANTRA

4.2.7. Titanic Dataset Analysis and Data Clea... 04:37

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ti
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thay
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",fe
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.0
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.86
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,34990
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,
```

Note: Refer to the visible test case for better reference.

Sample Test Cases +

Explorer | titanicData... | Submit | Debugger

```python
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']
data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# 1. Calculate the survival rate by class
print(data.groupby('Pclass')['Survived'].mean())
# 2. Calculate the survival rate by embarked location
print(data.groupby('Embarked_S')['Survived'].mean())
# 3. Calculate the survival rate by family size
print(data.groupby('FamilySize')['Survived'].mean())
# 4. Calculate the survival rate by being alone
print(data.groupby('IsAlone')['Survived'].mean())
# 5. Get the average fare by class
print(data.groupby('Pclass')['Fare'].mean())
# 6. Get the average age by class
print(data.groupby('Pclass')['Age'].mean())
# 7. Get the average age by survival status
print(data.groupby('Survived')['Age'].mean())
# 8. Get the average fare by survival status
print(data.groupby('Survived')['Fare'].mean())
# 9. Get the number of survivors by class
print(data[data['Survived'] == 1]['Pclass'].value_counts())
# 10. Get the number of non-survivors by class
print(data[data['Survived'] == 0]['Pclass'].value_counts())
```

CODETANTRA

4.2.8. Titanic Dataset Analysis and Data Clea... 06:45

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ti
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thay
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",fe
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.0
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.86
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,34990
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,
```

Note: Refer to the visible test case for better reference.

Sample Test Cases +

Explorer

titanicData...

Submit

Debugger

```python
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)


# 1. Get the number of survivors by gender
print(data[data['Survived'] == 1]['Sex'].value_counts())
# 2. Get the number of non-survivors by gender
print(data[data['Survived'] == 0]['Sex'].value_counts())
# 3. Get the number of survivors by embarked location
print(data[data['Survived'] == 1]['Embarked_S'].value_counts())
# 4. Get the number of non-survivors by embarked location
print(data[data['Survived'] == 0]['Embarked_S'].value_counts())
# 5. Calculate the percentage of children (Age < 18) who survived
print(data[data['Age']<18]['Survived'].mean())
# 6. Calculate the percentage of adults (Age >= 18) who survived
print(data[data['Age'] >= 18]['Survived'].mean())
# 7. Get the median age of survivors
print(data[data['Survived'] == 1]['Age'].median())
# 8. Get the median age of non-survivors
print(data[data['Survived'] == 0]['Age'].median())
# 9. Get the median fare of survivors
print(data[data['Survived'] == 1]['Fare'].median())
# 10. Get the median fare of non-survivors
print(data[data['Survived'] == 0]['Fare'].median())
```